

- Seasonal start-up date
- Occupant move-in date
- Operator training
- Supporting a process, such as operation of computer equipment
- Promoting a germ-free environment
- Increasing marketability of rental spaces
- Increasing net rental income
- Increasing property saleability
- Public image of the property
- Certification of energy use (LEED®, ENERGYSTAR®, etc.)

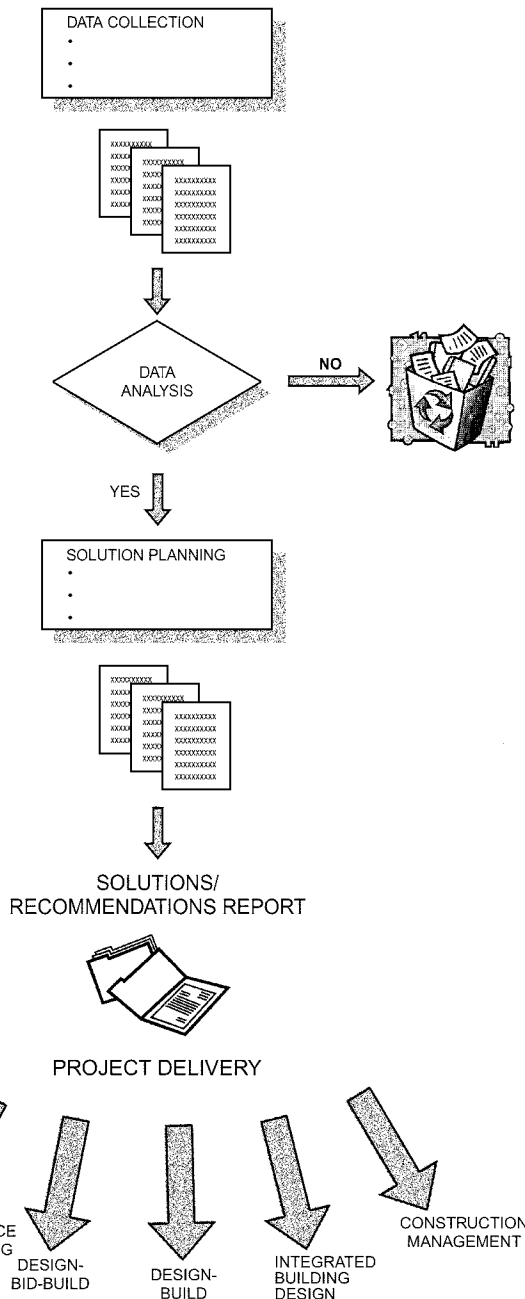


Fig. 1 Process Flow Diagram
(Courtesy RDK Engineers)

The owner can only make appropriate value judgments if the design engineer provides complete information on the advantages and disadvantages of each system option. Just as the owner does not usually know the relative advantages and disadvantages of different HVAC systems, the design engineer rarely knows all the owner's financial and functional goals. Hence, the owner must be proactive and involved in system selection in the conceptual phase of the job. The same can be said for operator participation so that the final design intent is sustainable and can continue to fulfill design intent.

All owners should request and/or require the design team to provide building and HVAC security (as defined in Chapter 61 of the 2019 *ASHRAE Handbook—HVAC Applications*). This security, in addition to environmental health and safety criteria, should address outside-the-building influences (e.g., smoke, toxic fumes, flood, etc.) as well as influences inside the building. System selection pertaining to HVAC security may require another design team member, a security consultant, and/or the owner's own security group.

Equipment and System Constraints

Once the goal criteria and additional goal options are verified, equipment and system constraints must be assessed and documented as part of the system analysis and selection recommendation report. These constraints may include the following:

- Performance limitations (e.g., temperature, humidity, space pressure, etc.)
- Code updates and/or new codes
- Available capacity including equipment size, as well as ductwork and/or pipe size
- Available space and access in and/or out with new or old equipment
- Available utility source
- Energy budget (e.g., conceptual Btu/h-ft² per year), code-driven or targeted
- Equipment efficiency
- Operator knowledge and capabilities
- Existing building occupants
- Building architecture

Few projects allow detailed quantitative evaluation of all alternatives. Common sense, historical data, and subjective experience can be used to narrow choices to one or two potential systems. Heating and air-conditioning loads often contribute to constraints, narrowing the choice to systems that fit in available space and are compatible with building architecture. Chapters 17 and 18 of the 2017 *ASHRAE Handbook—Fundamentals* describe methods to determine the size and characteristics of heating and air-conditioning loads. By establishing the capacity requirement, equipment size can be determined, and the choice may be narrowed to those systems that work well on projects within the required size range.

Loads vary over time based on occupied and unoccupied periods, changes in weather, type of occupancy, activities, internal loads, and solar exposure. Each space with a different use and/or exposure may require its own control zone to maintain space comfort. Some areas with special requirements (e.g., dehumidification requirements) may need individual systems. The extent of zoning, degree of control required in each zone, and space required for individual zones also narrow system choices.

No matter how efficiently a particular system operates or how economical it is to install, it can only be considered if it (1) maintains the desired building space environment within an acceptable tolerance under expected conditions and occupant activities and (2) physically fits into, on, or adjacent to the building without causing objectionable occupancy conditions.

Cooling, heating, and humidity control are often the basis of sizing HVAC components and subsystems, but ventilation requirements can also significantly impact system sizing. For example, if

Duct Insulation

In new construction and renovation upgrade projects, HVAC supply air ducts should be insulated and sealed in accordance with energy code requirements. ASHRAE *Standard* 90.1 and Chapter 23 of the 2017 ASHRAE *Handbook—Fundamentals* have more information about insulation and calculation methods.

Ceiling and Floor Plenums

Frequently, the space between the suspended ceiling and the floor slab above it is used as a return air plenum to reduce distribution ductwork. Check regulations before using this approach in new construction or renovation because most codes prohibit combustible material in a ceiling return air plenum. Ceiling plenums and raised floors can also be used for supply air displacement systems to minimize horizontal distribution, along with other features discussed in Chapter 4.

Some ceiling plenum applications with lay-in panels do not work well where the stack effect of a high-rise building or high-rise elevators creates a negative pressure. If the plenum leaks to the low-pressure area, tiles may lift and drop out when the outside door is opened and closed.

The air temperature in a return air plenum directly below a roof deck is usually higher by 3 to 5°F during the air-conditioning season than in a ducted return. This can be an advantage to the occupied space below because heat gain to the space is reduced. Conversely, return air plenums directly below a roof deck have lower return air temperatures during the heating season than a ducted return and may require supplemental heat in the plenum.

Using a raised floor for air distribution is popular for computer rooms and cleanrooms, and is now being used in other HVAC applications. Underfloor air displacement (UFAD) systems in office buildings use the raised floor as a supply air plenum, which could reduce overall first cost of construction and ongoing improvement costs for occupants. Making the underfloor supply air plenum as airtight as possible is critical to the success of this type of plenum. This UFAD system improves air circulation to the occupied area of the space. See Chapter 20 of the 2019 ASHRAE *Handbook—HVAC Applications* and Chapter 20 of the 2017 ASHRAE *Handbook—Fundamentals* for more information on displacement ventilation and underfloor air distribution.

5. PIPE DISTRIBUTION

Piping should deliver refrigerant, heating water, chilled water, condenser water, fuel oil, gas, steam, and condensate drainage and return to and from HVAC equipment as directly, quietly, and economically as possible. Structural features of the building generally require mechanical and electrical coordination to accommodate P-traps, pipe pitch-draining of low points in the system, and venting of high points. When assessing application of pipe distribution to air distribution, the floor-to-floor height requirement can influence the pipe system: it requires less ceiling space to install pipe when compared to air ductwork. An alternative to horizontal piping is vertical pipe distribution, which may further reduce floor-to-floor height criteria. See Chapter 22 of the 2017 ASHRAE *Handbook—Fundamentals* for pipe sizing.

Pipe Systems

HVAC piping systems can be divided into two parts: (1) piping in the central plant equipment room and (2) piping required to deliver refrigerant, heating water, chilled water, condenser water, fuel oil, gas, steam, and condensate drainage and return to and from decentralized HVAC and process equipment throughout the building. Chapters 11 to 15 discuss piping for various heating and cooling systems. Chapters 1 to 4 of the 2018 ASHRAE *Handbook—Refrigeration* discuss refrigerant system design and practices.

Pipe Insulation

In new construction and renovation projects, most HVAC piping must be insulated. ASHRAE *Standard* 90.1 and Chapter 23 of the 2017 ASHRAE *Handbook—Fundamentals* have information on insulation and calculation methods. In most applications, water-source heat pump loop piping and condenser water piping may not need to be insulated.

6. SECURITY AND ENVIRONMENTAL HEALTH AND SAFETY

Since September 11, 2001, much attention has been given to protecting buildings against terrorist attacks via their HVAC systems. The first consideration should be to perform a risk assessment of the particular facility, which may be based on usage, size, population, and/or significance. The risk assessment is a subjective judgment by the building owner (and sometimes by a registered/certified security professional) of whether the building is at low, medium, or high risk. An example of low-to-medium risk buildings may be suburban office buildings, shopping malls, hospitals, educational institutions, or major office buildings. High-risk buildings may include major government buildings. The level of protection designed into these buildings may include enhanced particulate filtration, gaseous-phase filtration, and various control schemes to allow purging of the facility using either the HVAC system or an independent dedicated system.

Enhanced particulate filtration for air-handling systems to the level of MERV 14 to 16 filters not only tends to reduce circulation of dangerous substances (e.g., anthrax), but also provides better indoor air quality (IAQ). Gaseous-phase filtration can remove harmful substances such as sarin and other gaseous threats. For buildings of all risk levels, consideration should be taken to include proper location of outdoor air intakes to account for other security concerns, such as toxic fumes from a local accident (e.g., punctured tank from a nearby railroad car or tanker truck). The extent to which the HVAC system designer should use these measures could depend on the recommendations of an experienced security and environmental health and safety consultant or agency.

In any building, consideration should be given to protecting outdoor air intakes against insertion of dangerous materials by locating the intakes on the roof or substantially above grade level. Separate systems for mailrooms, loading docks, and other similar spaces should be considered so that any dangerous material received cannot be spread throughout the building from these types of vulnerable spaces. Emergency ventilation systems for these types of spaces should be designed so that upon detection of suspicious material, toxic fumes, or chemical spill, these spaces can be quickly purged and maintained at a negative pressure.

A more extensive discussion of this topic can be found in ASHRAE's *Guideline* 29. Note, however, that fan energy required to move air through these high efficiency filters is very significant.

7. AUTOMATIC CONTROLS AND BUILDING MANAGEMENT SYSTEMS

Basic HVAC system controls are primarily direct digital controls (DDC), but there are also electric, electronic, and pneumatic controls (found largely as part of existing building HVAC systems). Depending on the application, the design engineer may recommend a simple and basic system strategy as a cost-effective solution to an owner's heating, ventilation, and cooling needs. Chapter 48 of the 2019 ASHRAE *Handbook—HVAC Applications* and Chapter 7 of the 2017 ASHRAE *Handbook—Fundamentals* discuss automatic control in more detail.

The next level of HVAC system management is DDC wireless systems. Many DDC installations will use either pneumatic or

units, and snow-melting panels. A large storage tank may be included in the system to store energy for use when heat input devices such as the boiler or a solar energy collector are not supplying energy.

This chapter covers the principles and procedures for designing and selecting piping and components for low-temperature water, chilled water, and dual-temperature water systems. See [Chapter 14](#) for information on medium- and high-temperature water systems.

2. CLOSED WATER SYSTEMS

Because most hot- and chilled-water systems are closed, this chapter addresses only closed systems. The fundamental difference between a closed and an open water system is the interface of the water with a compressible gas (such as air) or an elastic surface (such as a diaphragm). A **closed water system** is defined as one with no more than one point of interface with a compressible gas or surface, and that will not create system flow by changes in elevation. This definition is fundamental to understanding the hydraulic dynamics of these systems. Earlier literature referred to a system with an open or vented expansion tank as an “open” system, but this is actually a closed system; the atmospheric interface of the tank simply establishes the system pressure.

An **open system**, on the other hand, has more than one such interface. For example, a cooling tower system has at least two points of interface: the tower basin and the discharge pipe or nozzles entering the tower. One major difference in hydraulics between open and closed systems is that some hydraulic characteristics of open systems cannot occur in closed systems. For example, in contrast to the hydraulics of an open system, in a closed system (1) flow cannot be motivated by static head differences, (2) pumps do not provide static lift, and (3) the entire piping system is always filled with water.

[Figure 1](#) shows the fundamental components of a closed hydronic system. Actual systems generally have additional components such as valves, vents, regulators, etc., but these are not essential to the basic principles underlying the system.

These fundamental components are

- Loads
- Source
- Expansion chamber
- Pump
- Distribution system

Theoretically, a hydronic system could operate with only these five components.

The components are subdivided into two groups: thermal and hydraulic. Thermal components consist of the load, source, and expansion chamber. Hydraulic components consist of the distribution system, pump, and expansion chamber. The expansion chamber is the only component that serves both a thermal and a hydraulic function.

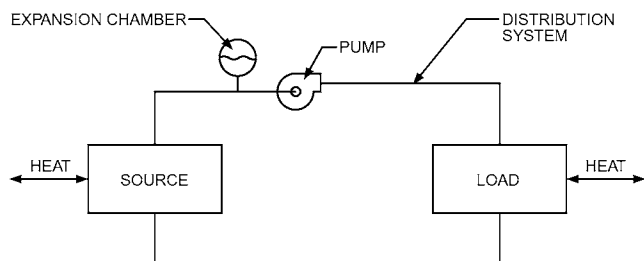


Fig. 1 Fundamental Components of Hydronic System

2.1 METHOD OF DESIGN

This section outlines general steps a designer may follow to complete system design. The methodology is not a rigid framework, but rather a flexible outline that should be adapted by the designer to suit current needs. The general order as shown is approximately chronological, but note that succeeding steps often affect preceding steps, so a fundamental reading of this entire chapter is required to fully understand the design process.

1. **Determine system and zone loads.** Loads are covered in Chapters 14 to 19 of the 2017 *ASHRAE Handbook—Fundamentals*. Several load calculation procedures have been developed, with varying degrees of calculation accuracy. The load, in conjunction with design water temperature difference, determines the flow of the hydronic system, which ultimately affects the system's heat transfer ability and energy performance. Designers should apply the latest computerized calculation methods for optimal system design. Load calculation should also detail the facility's loading profile facility to enhance the hydronic system control strategy.
2. **Select comfort or process heat transfer devices.** This often means a coil- or water-to-air heat exchanger (terminal). Coil selection and operation has the single largest influence on hydronic system design. Coils implement the design criteria of flow, temperature difference, and control ability. Coil head loss and location affects pipe design and sizing, control devices, and pump selection. For details on coils, see [Chapters 23](#) and [27](#).
3. **Select system distribution style(s).** Based on the load and its location, different piping styles may be appropriate for a given design. Styles may be comingled in a successful hydronic system design to optimize building performance. Schematically lay out the system to establish a preliminary design.
4. **Size branch piping system.** Based on the selection of the coil, its controlling devices, style of installation, and location, branch piping is sized to provide required flow, and head loss is calculated.
5. **Calculate distribution piping head loss.** Although the criteria for pipe selection in branch and distribution system piping may be similar, understanding the relationship and effect of distribution system head loss is important in establishing that all terminals get the required flow for the required heat transfer.
6. **Lay out piping system and size pipes.** After preliminary calculations of target friction loss for the pipes, sketch the system. After the piping system is laid out and the calculations of actual design head loss are complete, note the losses on the drawings for the commissioning process.
7. **Select pump specialties.** Any devices required for operation or measurement are identified, so their head loss can be determined and accounted for in pump selection.
8. **Select air management methodology.** All hydronic systems entrain air in the circulated fluid. Managing the collection of that air as it leaves the working fluid is essential to management of system pressure and the safe operation of system components.
9. **Select pump (hydraulic components).** Unless a system is very small (e.g., a residential hot-water heating system), the pump is selected to fit the system. A significant portion of energy use in a hydronic system is transporting the fluid through the distribution system. Proper pump selection limits this energy use, whereas improper selection leads to energy inefficiency and poor distribution and heat transfer.
10. **Determine installation details, iterate design.** Tuning the design to increase performance and cost effectiveness is an important last step. Documenting installation details is also important, because this communication is necessary for well-built designs and properly operated systems.

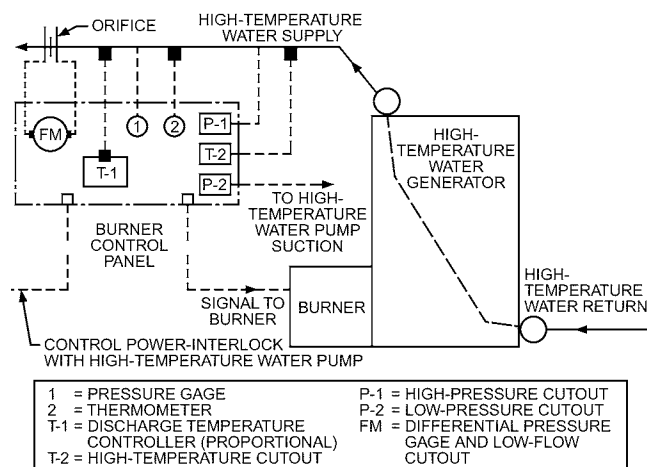


Fig. 9 Control Diagram for HTW Generator

reflect the heat downward and to prevent smudging the surfaces above the pipe.

8. INSTRUMENTATION AND CONTROLS

Pressure gages should be installed in the pump discharge and suction and at locations where pressure readings will assist operation and maintenance. Thermometers (preferably dial-type) or thermometer wells should be installed in the flow and return pipes, the pump discharge, and at any other points of major temperature change or where temperatures are important in operating the system. It is desirable to have thermometers and gages in the piping at the entrance to each building converter.

Inert-gas-pressurized systems should be controlled from the generator discharge temperature. Combustion controls are discussed in detail in [Chapter 31](#).

In the water-tube generators most commonly used for HTW applications, the flow of water passes through the generator in seconds. The temperature controller must have a rapid response to maintain a reasonably uniform leaving water temperature.

Keep the controls simple. The rapid response through the generator makes it necessary to modulate the combustion rate on all systems with a capacity of over a few million Btu/h. In the smaller size range, this can be done by high/low firing. In large systems, particularly those used for central heating applications, full modulation of the combustion rate is desirable through at least 20% of full capacity. On/off burner control is generally not used in steam-pressurized cycles because the system loses pressurization during the off cycle, which can cause flashing and cavitation at the HTW pumps.

All generators should have separate safety controls to shut down the combustion apparatus when the system pressure or water temperature is high. HTW generators require a minimum water flow at all times to prevent tube failure. Means should be provided to measure the flow and to stop combustion if the flow falls below the minimum value recommended by the generator manufacturer. For inert-gas-pressurized cycles, a low-pressure safety control should be included to shut down the combustion system if pressurization is lost. [Figure 9](#) shows the basic schematic control diagram for an HTW generator.

Control valve selection and sizing are important because of relatively high temperature drops and smaller flows in HTW systems. The valve must be sized so that it is effective over its full range of travel. The valve and equipment must be sized to absorb, in the control valve at full flow, not less than half the available pressure dif-

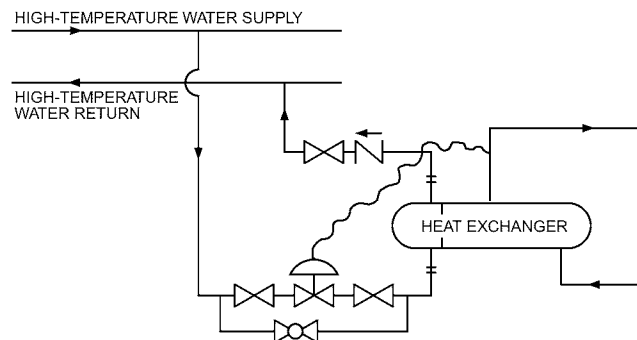


Fig. 10 Heat Exchanger Connections

ference between supply and return mains where the equipment is served. An equal-percentage valve is typically used. Sometimes, two small valves or a one-third/two-third provide better control than one large valve, due to the turndown capabilities over the range of flow. Stainless steel trim is recommended, and all valve body materials and packing should be suitable for high temperatures and pressures. The valve should have a close-off rating at least equal to the maximum pressure produced by the circulating pump. Generally, two-way valves are more desirable than three-way valves because of equal percentage flow characteristics and the smaller capacities available in two-way valves. Single-seated valves are preferable to double-seated valves because the latter do not offer tight shutoff capability.

Control valves are commonly located in the return lines from heat transfer units to reduce the valve operating temperature and to prevent plug erosion caused by high-temperature water flashing to steam at lower discharge pressure. A typical application is for controlling the temperature of water being heated in a heat exchanger where the heating medium is high-temperature water. The temperature-measuring element of the controller is installed on the secondary side and should be located where it can best detect changes to prevent overheating of outlet water. When the measuring element is located in the outlet pipe, there must be a continuous flow through the exchanger and past the element. The controller regulates the HTW supply to the primary side by means of the control valve in the HTW return. If the water leaving the exchanger is used for space heating, the set point of the thermostat in this water can be readjusted according to outdoor temperature.

Another typical application is to control a low- or medium-pressure steam generator, usually less than 50 psig, using high-temperature water as the heat source. In this application, a proportional pressure controller measures the steam pressure on the secondary side and positions the HTW control valve on the primary side to maintain the desired steam pressure. For general information on automatic controls, refer to Chapter 48 of the 2019 *ASHRAE Handbook—HVAC Applications*.

Where submergence is sufficient to prevent flashing in the **vena contracta**, control valves can be in the HTW supply instead of in the return to water heaters and steam generators ([Figure 10](#)). When used in conjunction with a check valve in the return, this arrangement shuts off the high-temperature water supply to the heat exchangers if a tube bundle leaks or ruptures.

9. WATER TREATMENT

Water treatment for HTW systems should be referred to a specialist. However, the design engineer should be conversant with the technology to ensure that systems are not overtreated. MTW/HTW systems designed to minimize water loss in the system require minimal chemical additions. Experience shows that, in a well-designed,

Table 1 Typical Applications for Supply Air Outlets

Outlet Types	Fully Mixed			Fully Stratified		Partially Mixed		
	Ceiling Mounted	Wall Mounted	Floor/Sill	Wall Mounted	Floor/Sill	Ceiling Mounted	Wall Mounted	Floor/Sill
Grilles								
Adjustable blade	⊙	●	⊗	⊗	⊗	⊗	⊙	⊙
Fixed blade	⊙	⊙	⊗	⊙	⊙	⊗	⊙	⊗
Linear bar	⊗	●	●	⊙	⊙	⊗	⊙	●
Nozzle and drum louver	⊙	●	⊗	⊗	⊗	⊗	⊗	⊗
Diffusers								
Round	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Square	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Perforated face	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Louvered face	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Plaque face	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Hemispherical	⊙	⊗	⊗	⊗	⊗	⊙	⊗	⊗
Laminar flow	⊙	⊗	⊗	⊗	⊗	⊙	⊗	⊗
Linear slot	●	●	⊗	⊗	⊗	⊗	⊗	⊗
T-bar slot	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Light troffer	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
Swirl	●	⊗	⊗	⊗	⊙	⊗	⊙	●
Displacement	⊗	⊗	⊗	●	●	⊙	⊗	⊗
Air dispersion duct	●	⊙	⊗	⊗	⊗	⊙	⊙	⊗

● = often used ⊙ = sometimes used ⊙ = seldom used ⊗ = not recommended

grille can be set for approximately 50° maximum deflection to either side, which can usually cover the conditioned space. Horizontal blades can be set to control the elevation of the discharge pattern. Upward deflection minimizes thermal drop in cooling applications, especially when the air jet connects to a ceiling surface.

Ceiling. Such installation is generally limited to ceiling diffusers (or grilles with curved vanes) that discharge parallel to the mounting surface. For high mounting locations (greater than 10 ft above the occupied zone), vertical discharge may be the preferred application. Supply outlets installed in 8 to 10 ft high ceilings can achieve satisfactory performance if appropriate consideration is made for terminal velocities and temperature differentials in the occupied space. Discharging the airstream directly into occupied zone may result in unacceptable comfort conditions. Supply outlet selections for heating and cooling applications from the same device should be carefully examined for use in both applications.

Sill. Linear bar grilles are commonly used in sill applications. The grille should be installed with the supply air jet directed vertically away from the occupied space. When the device is mounted 12 in. or less from the wall, a device with 0° deflection is suitable. If the device is installed more than 12 in. from the vertical surface, a linear bar grille with a fixed 15 to 30° deflection is recommended. This device should be installed with the jet directed toward the wall. These grilles can be available with doors, removable cores, or other means of access to mechanical equipment that may be installed in the sill enclosure. The presence of window draperies or blinds and the effect of an impinging airstream must be considered in the selection.

Floor. Linear bar grilles are often used for floor applications. The designer should determine the traffic and floor loading on the grille (consult local building code for minimum rating requirements) and consult the manufacturer's load limit for the grille. The grille should be placed in low-traffic areas. A floor-mounted grille is usually selected to discharge supply air along a wall or exterior surface. Install with the supply air jet directed vertically away from the occupied space. When the device is mounted 12 in. or less from the wall, a device with 0° deflection is suitable. If the device is installed more than 12 in. from the vertical surface, a linear bar grille with a fixed

15 to 30° deflection is recommended. This device should be installed with the jet directed toward the wall. The presence of window draperies or blinds and the effect of an impinging airstream must be considered in the selection.

Underfloor air distribution (UFAD) typically employs swirl or linear bar grille type supply outlets recessed into the floor that discharge directly into the occupied zone. When sizing and locating the outlets care should be taken to assure occupant thermal comfort. Displacement ventilation typically employs supply outlets fixed to the wall, sometimes mounted in the ceiling or floor, that discharge directly into the occupied zone. When sizing and locating the outlets, care should be taken to ensure occupant thermal comfort.

Table 1 shows the types of supply air outlets and provides guidance for best use practices. This table is for guidance only; designers should consult manufacturers' literature for additional application information.

3.2 RETURN AND EXHAUST AIR INLETS

Return air inlets may either be connected to a duct or simply cover openings that transfer air from one area to another. Exhaust air inlets remove air directly from a building and, therefore, are most always connected to a duct. Velocity, sound, and pressure requirements for inlets are determined by size and configuration. In general, the same type of equipment, grilles, slot diffusers, and ceiling diffusers used for supplying air may also be used for air return and exhaust. However, inlets do not require the deflection, flow equalizing, and turning devices necessary for supply outlets. Return and exhaust inlets may be mounted in practically any location (e.g., ceilings, high or low sidewalls, floors, and doors). Dampers such as the one shown in Figure 3A are sometimes used in conjunction with grille return and exhaust inlets to aid system balancing. When used in a partially mixed or fully stratified system, the returns/exhaust outlets should be located above the occupied zone and either ceiling or high sidewall mounted (unless superseded by other codes or standards).

mance of coils at various air and water velocities and entering humidity and temperatures. Most manufacturers provide computerized coil selection programs to potential customers. The final choice can then be made based on system performance and economic requirements.

Performance and Ratings

The long-term performance of an extended-surface air-cooling and dehumidifying coil depends on its correct design to specified conditions and material specifications, proper matching to other system components, proper installation, and proper maintenance as required.

In accordance with AHRI *Standard* 410, dry-surface (sensible cooling) coils and dehumidifying coils (which both cool and dehumidify), particularly those for field-assembled coil banks or factory-assembled packaged units using different combinations of coils, are usually rated within the following parameters:

Entering air dry-bulb temperature: 65 to 100°F

Entering air wet-bulb temperature: 60 to 85°F (if air is not dehumidified in the application, select coils based on sensible heat transfer)

Air face velocity: 200 to 800 fpm

Evaporator refrigerant saturation temperature: 30 to 55°F at coil suction outlet (refrigerant vapor superheat at coil suction outlet is 6°F or higher)

Entering chilled-water temperature: 35 to 65°F

Water velocity: 1 to 8 fps

For ethylene glycol solution: 1 to 6 fps, 0 to 90°F entering dry-bulb temperature, 60 to 80°F entering wet-bulb temperature, 10 to 60% aqueous glycol concentration by weight

The air-side ratio of sensible to total heat removed by dehumidifying coils varies in practice from about 0.6 to 1.0 (i.e., sensible heat is from 60 to 100% of the total, depending on the application). For information on calculating a dehumidifying coil's sensible heat ratio, see the section on Performance of Dehumidifying Coils, or Appendix D of *Cooling and Heating Load Calculation Principles* (Pedersen et al. 1998). For a given coil surface design and arrangement, the required sensible heat ratio may be satisfied by wide variations in and combinations of air face velocity, in-tube temperature, flow rate, entering air temperature, coil depth, etc., although the variations may be self limiting. The maximum coil air face velocity should be limited to a value that prevents water carryover into the air ductwork. Dehumidifying coils for comfort applications are frequently selected in the range of 400 to 500 fpm air face velocity.

Operating ratings of dehumidifying coils for factory-assembled, self-contained air conditioners are generally determined in conjunction with laboratory testing for the system capacity of the complete unit assembly. For example, a standard rating point has been 33.4 cfm per 1000 Btu/h (or 400 cfm per ton of refrigeration effect), not to exceed 37.5 cfm per 100 Btu/h for unitary equipment. Refrigerant (e.g., R-22) duty would be 6 to 10°F superheat for an appropriate balance at 45°F saturated suction. For water coils, circuitry would operate at 4 fps, 42°F inlet water, 12°F rise (or 2 gpm per ton of refrigeration effect). The standard ratings at 80°F db and 67°F wb are representative of the entering air conditions encountered in many comfort operations. Although indoor conditions are usually lower than 67°F wb, it is usually assumed that introducing outdoor air brings the air mixture to the cooling coil up to about 80°F db/67°F wb entering air design conditions.

Dehumidifying coils for field-assembled projects and central station air-handling units were formerly selected according to coil rating tables but are now selected by computerized selection programs. Either way, selecting coils from the load division indicated by the load calculation works satisfactorily for the usual human comfort applications. Additional design precautions and refinements are necessary for more exacting industrial applications and for all types

of air conditioning in humid areas. One such refinement, the dual-path air process, uses a separate cooling coil to cool and dehumidify ventilation air before mixing it with recirculated air. This process dehumidifies what is usually the main source of moisture: makeup outdoor air. Condenser heat reclaim (when available) is another refinement required for some industrial applications and is finding greater use in commercial and comfort applications.

Airflow ratings are based on standard air of 0.075 lb/ft³ at 70°F and a barometric pressure of 29.92 in. Hg. In some mountainous areas with a sufficiently large market, coil ratings and altitude-corrected psychrometrics are available for their particular altitudes.

When checking the operation of dehumidifying coils, climatic conditions must be considered. Most problems are encountered at light-load conditions, when the cooling requirement is considerably less than at design conditions. In hot, dry climates, where the outdoor dew point is consistently low, dehumidifying is not generally a problem, and the light-load design point condition does not pose any special problems. In hot, humid climates, the light-load condition has a higher proportion of moisture and a correspondingly lower proportion of sensible heat. The result is higher dew points in the conditioned spaces during light-load conditions unless a special means for controlling inside dew points (e.g., reheat or dual path) is used.

Fin surface freezing at light loads should be avoided. Freezing occurs when a dehumidification coil's surface temperature falls below 32°F. Freezing does not occur with standard coils for comfort installations unless the refrigerant evaporating temperature at the coil outlet is below 25 to 28°F saturated; the exact value depends on the design of the coil, its operating dew point, and the amount of loading. With coil and condensing units to balance at low temperatures at peak loads (not a customary design choice), freezing may occur when load suddenly decreases. The possibility of this type of surface freezing is greater if a bypass is used because it causes less air to be passed through the coil at light loads.

4. AIRFLOW RESISTANCE

A cooling coil's airflow resistance (air friction) depends on the tube pattern and fin geometry (tube size and spacing, fin configuration, and number of in-line or staggered rows), coil face velocity, and amount of moisture on the coil. The coil air friction may also be affected by the degree of aerodynamic cleanliness of the coil core; burrs on fin edges may increase coil friction and increase the tendency to pocket dirt or lint on the faces. A completely dry coil, removing only sensible heat, offers approximately one-third less resistance to airflow than a dehumidifying coil removing both sensible and latent heat.

For a given surface and airflow, increasing the number of rows or fins increases airflow resistance. Therefore, final selection involves economic balancing of the initial cost of the coil against the operating costs of the coil geometry combinations available to adequately meet the performance requirements.

The aluminum fin surfaces of new dehumidifying coils tend to inhibit condensate sheeting action until they have aged for a year. Hydrophilic aluminum fin surface coatings reduce water droplet surface tension, producing a more evenly dispersed wetted surface action at initial start-up. Manufacturers have tried different methods of applying such coatings, including dipping the coil into a tank, coating the fin stock material, or subjecting the material to a chemical etching process. Tests have shown as much as a 30% reduction in air pressure drop across a hydrophilic coil as opposed to a new untreated coil.

5. HEAT TRANSFER

The heat transmission rate of air passing over a clean tube (with or without extended surface) to a fluid flowing in it is impeded principally by three thermal resistances: (1) surface air-side film thermal